ETH zürich



AML 2026 – Practical Projects – Task 1

Brain Age Prediction from MRI Features

Introduction tutorial · 19 October 2026

Johannes Schulte-Vels
jschul@ethz.ch

Nikolaj Murasov
nmurasov@ethz.ch

Agenda

1	**Background**	MRI and brain aging
2	**The Task**	Three pre-processing subtasks and age regression
3	**Data**	832 MRI features, files and download
4	**Evaluation**	Coefficient of determination R^2
5	**Approaches**	A simple baseline pipeline and tips
6	**Logistics & Grading**	Timeline, hand-in, grading, rules
7	**FAQ & Contact**	Frequently asked questions, who to ask

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Magnetic Resonance Imaging (MRI)

- Uses a strong magnetic field and radio waves to produce detailed 3D anatomical images
- Non-invasive and non-radiative: well suited to investigate sensitive organs such as the brain
- Produces high-resolution images
- MRI machines are large, tube-shaped magnets
- A key technology in medical imaging, but unevenly available across the world



45.0

Switzerland (2016)
MRI units per million inhabitants

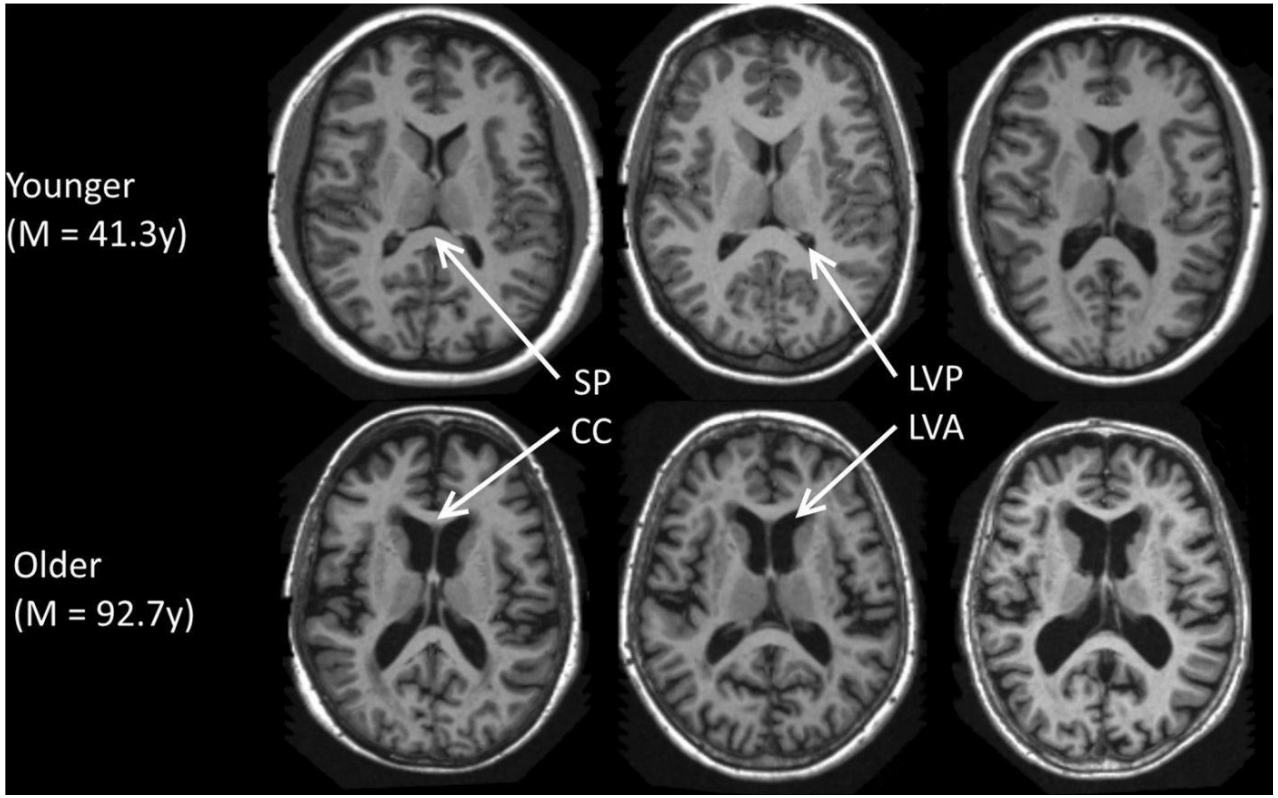
0.22

West Africa (2018)
MRI units per million inhabitants

Source: OECD, Magnetic resonance imaging (MRI) units

Aging Effect on Brain MRI

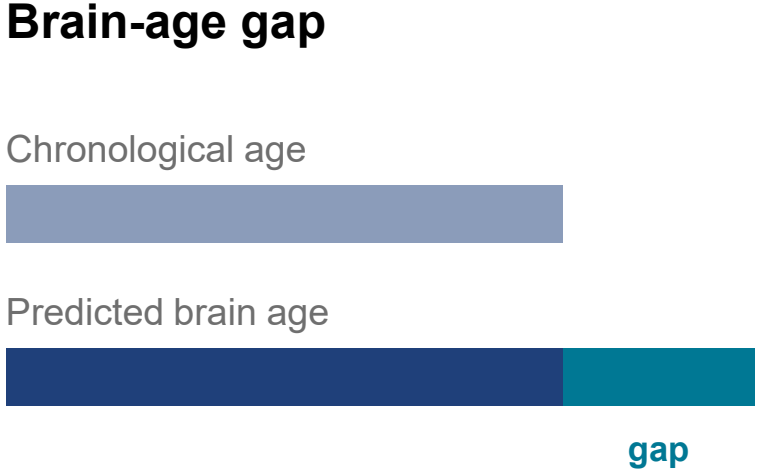
- The brain undergoes profound age-related neuroanatomical changes during the aging process
- The global grey matter volume decreases with age
- Grey matter contains most of the brain's neuronal cell bodies



Davis, N. J. "Brain stimulation for cognitive enhancement in the older person: State of the art and future directions." Journal of Cognitive Enhancement 1.3 (2017): 337–344.

Brain Age Estimation: Why Is It Interesting?

- **Aging does not affect people uniformly!**
- Individual rates of aging are shaped by interactions between environmental, genetic and epigenetic factors
- Brain MRI studies show a relation between accelerated aging and accelerated brain atrophy
- Brain age could improve early diagnosis and risk assessment for age-associated neurodegenerative and neuropsychiatric diseases at a subject level
  - Alzheimer's, Parkinson's, Huntington's, …



Predicted brain age vs. chronological age.
A positive gap can indicate accelerated brain aging.

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Task 1 at a Glance

Input

- 832 anatomical features per subject, derived from brain MRI
- Contains missing values, outliers and irrelevant features

→

Your pipeline

- Impute missing values
- Detect and remove outliers
- Select features
- Train a regressor

→

Output

- The predicted age of every test subject
- One value per row, in the format of sample.csv

Scored by **R^2** between your predictions and the true ages of the test subjects (public and private part of the test set).

AML Task 1: Age Prediction

We have modified the derived input data in three ways:

1 **Irrelevant features**
Features that carry no information about age

2 **Outliers**
Samples in the training set that do not follow the data

3 **Perturbations**
E.g. missing values (NaN) in training and test set

	A	B	C	D	E	F	G	H	I	J
1	id	x0	x1	x2	x3	x4	x5	x6	x7	x8
2	0	14168.82	10514.38	3316.15	94230.7	102.3866	92.67713	11108.75	10866.51	10837.62
3	1	17757.04		4101.016	92959.53		99.85517	10013.96	10826.61	10076.1
4	2	14226.66	11029.64		124055.6	100.5425	92.86089		10492.34	
5	3	8766.012	7384.203	2147.308	100157.7	104.8551	101.929	10050.05	10499.52	10525.03
6	4	13801.02	13269.49	3408.317	92048.53	103.7598	95.78923	9667.354	10750.78	10618.8
7	5	11333.67	9693.98	2930.261	98892.45	103.9677	96.53505	10143.72	10889.1	10410.71
8	6	18012.99	13437.22	3928.789	105320.7	109.3164	102.4219	10218.4	10790.86	10252.3
9	7	17471.68	13219.53	3438.191		104.2403	99.32157	10678.33	10137.17	10860.43
10	8		11678.01	2782.414	93217.44	106.8652	106.655	10967.66	10236.58	10217.15
11	9	15821.05		4027.451	98841.36	103.7156	100.2716	11746.87	10549.48	10325.19
12	10	6480.594	6865.479	2074.261	99595.09	104.9383	101.421	10207.11	10520.72	10520.81
13	11	13190.9	9402.399	3224.518	84625.96	102.1383		8627.341	10362.81	10679.29
14	12	17182.58		3769.271	94769.97	108.786	113.3053	9836.233	10385.08	10041.71

Excerpt of X_train.csv: empty cells are missing values

Subtask 0: Filling Missing Values

Background

- There are missing values in the data; they are set to NaN
- Most methods cannot handle NaNs automatically
- Different strategies exist to impute missing values: mean, median, most frequent, …

What you need to do

- Fill in the missing values in the training set and in the test set

Subtask 1: Outlier Detection

Missing values → **Outlier detection** → Feature selection → Age regression

Background

- The training set contains outliers
- If the resulting model is not robust enough, it may be sensitive to the outliers
- Solution: outlier removal

What you need to do

- Build an outlier detection model that classifies every sample of the training set as outlier or not

Subtask 2: Feature Selection

Missing values → Outlier detection → **Feature selection** → Age regression

Background

- To make the task more challenging, we added manual features to the FreeSurfer-processed dataset

Advantages of feature selection

- Simpler, more interpretable models
- Shorter training times
- Better generalization by reducing overfitting

What you need to do

- Use feature selection methods to label every feature as selected or unselected
- Unselected features include irrelevant features and redundant features

Main Task: Age Prediction

Missing values → Outlier detection → Feature selection → **Age regression**

Background

- After pre-processing and dimensionality reduction, we finally arrive at the regression task

What you need to do

- Use suitable regression methods to predict the age of a person from the brain data
- Predict the age of every subject in X_test.csv: these predictions are what Kaggle scores

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MRI Processing

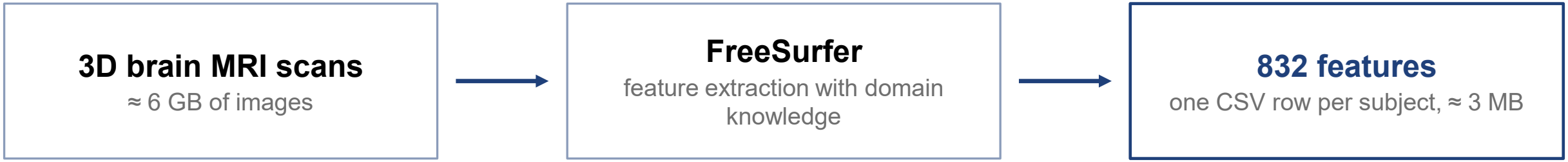
Raw brain scans are difficult to handle

- 3D brain scans have ~200 × 200 × 200 ≈ 10^7 voxels (1 voxel ≈ 1 mm^3)
- 3D structure and individual brain shapes make disease patterns difficult to recognize
- Data is scarce: a few thousand scans instead of millions of images

	ImageNet	Task 1 MRI data
Image size	224 × 224 × 3	~200 × 200 × 200
Data set size	1.2 million	~2 k

MRI Feature Extraction

We use 832 anatomical features for this project



Advantages

- No need to process big images, and no image analysis needed
- Meaningful features, e.g. cortex volume, left/right hemisphere surface area, white/grey matter volume

Disadvantage

- Information loss: whatever FreeSurfer does not measure is not in the data

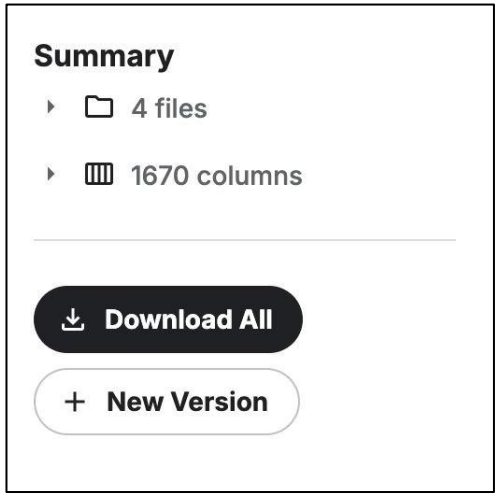
Files and How to Download Them

We provide the following files

- **X_train.csv, y_train.csv**: training set, features and labels
- **X_test.csv**: test set, make your predictions on this file
- **sample.csv**: a sample submission file in the correct format

Option A: Kaggle website

Competition page → Data → scroll down → “Download All” on the right



Option B: Kaggle API

```
$ pip install kaggle
# kaggle.com -> Settings -> API -> Create New Token
# (downloads kaggle.json)
$ mkdir -p ~/.kaggle
$ mv ~/Downloads/kaggle.json ~/.kaggle/
$ chmod 600 ~/.kaggle/kaggle.json
$ kaggle competitions download -c <competition>
```

<competition>: the competition name in the Kaggle URL (see Timeline & Key Facts)

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Evaluation Metric: Coefficient of Determination R^2

The proportion of the variance in the dependent variable that is predictable from the independent variables.

$$R^2 := 1 - \frac{SS_{res}}{SS_{tot}}$$

$$SS_{tot} := \sum_i (y_i - \overline{y})^2 \quad SS_{res} := \sum_i (f_i - y_i)^2$$

- Varies between 1 (best) and $-\infty$
- Predicting the mean $\bar{y}$ for every sample gives $R^2 = 0$
- R^2 is a scaled version of the MSE: unlike the MSE, it is invariant to scaling y

How to compute it in Python

```
from sklearn.metrics import r2_score
score = r2_score(y_true, y_pred)
```

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Baseline Pipeline: Loading the Data

Load data → Split & impute → Select features → Train & validate → Predict & submit

```
import numpy as np
import pandas as pd

X_train_df = pd.read_csv('./data/X_train.csv', skiprows=1, header=None)
y_train_df = pd.read_csv('./data/y_train.csv', skiprows=1, header=None)
X_test_df  = pd.read_csv('./data/X_test.csv',  skiprows=1, header=None)

# drop the id column
X_train = X_train_df.values[:, 1:]
y_train = y_train_df.values[:, 1:]
X_test  = X_test_df.values[:, 1:]
print(X_train.shape, y_train.shape, X_test.shape)
```

A deliberately simple starting point: it shows the mechanics, not a solution that passes the baseline.

Baseline Pipeline: Splitting and Imputing

Load data → **Split & impute** → Select features → Train & validate → Predict & submit

```
from sklearn.model_selection import train_test_split

# Randomly split the data into training and validation sets (80/20)
X_train, X_val, y_train, y_val = train_test_split(
    X_train, y_train, test_size=0.2, random_state=42)

# Impute missing values with the mean of each column (training statistics only)
X_mean  = np.nanmean(X_train, axis=0, keepdims=True)
X_train = np.where(np.isnan(X_train), X_mean, X_train)
X_val   = np.where(np.isnan(X_val),   X_mean, X_val)
X_test  = np.where(np.isnan(X_test),  X_mean, X_test)
```

Baseline Pipeline: Feature Selection

Load data → Split & impute → **Select features** → Train & validate → Predict & submit

```
from sklearn.feature_selection import SelectKBest, mutual_info_regression

# Select the 100 features with the highest mutual information with the age
selection = SelectKBest(mutual_info_regression, k=100).fit(X_train, y_train.ravel())
X_train = selection.transform(X_train)
X_val   = selection.transform(X_val)
X_test  = selection.transform(X_test)
```

Baseline Pipeline: Training and Validating

Load data → Split & impute → Select features → **Train & validate** → Predict & submit

```
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score

# Train a linear regression model
regressor = LinearRegression()
regressor.fit(X_train, y_train)

# Evaluate the model on the training and validation sets
train_score = r2_score(y_train, regressor.predict(X_train))
val_score   = r2_score(y_val,   regressor.predict(X_val))
print(train_score, val_score)
```

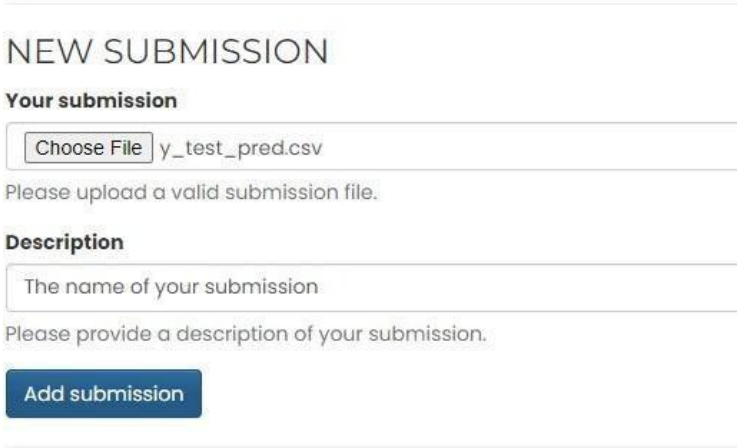
Note: no outlier detection yet, and a linear model. Expect a validation R^2 well below the baseline.

Baseline Pipeline: Export and Submit

Load data → Split & impute → Select features → Train & validate → **Predict & submit**

```
# Predict on the test set
y_test_pred = regressor.predict(X_test)

# Save the predictions in the format of sample.csv
table = pd.DataFrame({'id': np.arange(0, y_test_pred.shape[0]), 'y': y_test_pred.flatten()})
table.to_csv('./data/y_test_pred.csv', index=False)
```



- **On Kaggle:** Submit Prediction → upload y_test_pred.csv → add a short description.
- Then **select** the submission you want to be graded on.

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Project Timeline & Key Facts



Competition	https://www.kaggle.com/t/80f2dc3b47c442689ea3aa3f28a3fd4e
Evaluation metric	**R^2** (coefficient of determination), higher is better
Public baseline for grade 4.0	$R^2 = 0.5$

- All times are Zurich time. The public leaderboard is visible while the project is open; the private leaderboard is released when it closes.
- Only ETH accounts (ethz.ch, student.ethz.ch, inf.ethz.ch) can join the competition.
- Do not wait until the last day: servers get overloaded close to the deadline.

How to Hand In

1 Join Kaggle

Register with your ETH e-mail and form your team. Team names must be alphanumeric (A–Z, a–z, 0–9).

2 Submit predictions

Upload predictions in the format of sample.csv. Select the submission you want to be graded on.

3 Fill in the Moodle form

Every student individually: Kaggle team name, legi number (with hyphens), a description of your method, and your code.

- **A submission on Kaggle alone is not enough.** Without the Moodle form you receive no grade for the project.
- Write your description yourself. It must not be shared with your team members.
- Keep your code: your results must be reproducible, and we may ask you to reproduce them after the deadline.

How the Project is Graded

Three baselines per project

- $b_{1.0}$, $b_{4.0}$ and $b_{6.0}$ are fixed for each project. Grades are linearly interpolated between them.
- Your grade depends on your score only, not on your rank on the leaderboard.

From score to grade

- The **private** leaderboard score s is converted into a grade g.
- If $g \geq 4.0$, the project grade is g.
- If $g < 4.0$ but your **public** score reached $b_{4.0}$, the project grade is 4.0.
- Otherwise, the project grade is g.

Admission to the exam

Reach the 4.0 baseline on the public leaderboard in at least one of the two graded projects.

Project grade

The higher of your Task 1 and Task 2 grades counts, weighted 20 % in the final grade.

Do's and Don'ts

Do

- Handle missing values, outliers and irrelevant features before regressing
- Validate with cross-validation on the training data
- Describe what you did in the Moodle form
- Keep your implementation for a potential review
- Submit something early, then improve

Don't

- Use AutoML packages: anything that does automatic data cleaning or automatic model selection
- Overfit to the public leaderboard: it is only part of the test set
- Waste submissions: your team has a limited number of **10** per day
- Copy code or share your description with others

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Frequently Asked Questions

Q: Which programming language and tools am I supposed to use?

A: You are free to choose any programming language and any software library (except AutoML packages, see Do's and Don'ts).

Q: Can you give me a deadline extension?

A: We cannot grant deadline extensions, except in extraordinary cases (e.g. military service). We will require official confirmation of your problem (e.g. a medical certificate).

Q: Am I allowed to use ideas from published papers and use their code?

A: You may search for any material on how to train models for related tasks (papers, repositories, …). However, you must re-implement the code yourself. You are **not** allowed to copy code.

Frequently Asked Questions

Q: Can I post on Moodle as soon as I have a question?

A: Please first read the task description thoroughly, check this FAQ and discuss with your teammates. If you still need the TAs, post on the Moodle forum (or send a private message).

Q: When do we get the private scores and the grades?

A: The private leaderboard is released when the project closes. Grades are published on Moodle (legi number and grade only) once they have been validated.

Q: What if I do not reach the baseline in this task?

A: Admission to the exam requires reaching the 4.0 baseline on the public leaderboard in at least one of the two graded projects, so Task 2 still counts. The higher of your two project grades is used.

Who to Ask if I Have Questions?

In this order

1. **Moodle forum**
   General questions: others may have the same one
2. **Moodle private message**
   Private or non-public questions
3. **E-mail**
   Only if urgent or unanswered

Before asking: read the task description and the FAQ, and discuss with your teammates. Collaboration between teams beyond general discussion is not allowed.

Contacts

Task 1	Johannes Schulte-Vels jschul@ethz.ch Nikolaj Murasov nmurasov@ethz.ch
Course	Dr. Carlos Cotrini ccarlos@inf.ethz.ch
Forum	AML HS26 on Moodle moodle-app2.let.ethz.ch/mod/forum/view.php?id=1445686

Good luck!

Questions?